

Bioinformatics for engineers

Introduction to statistical inference

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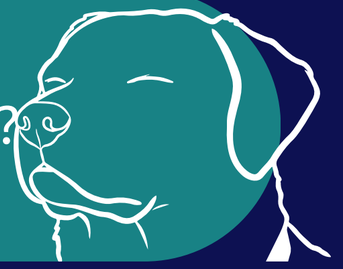
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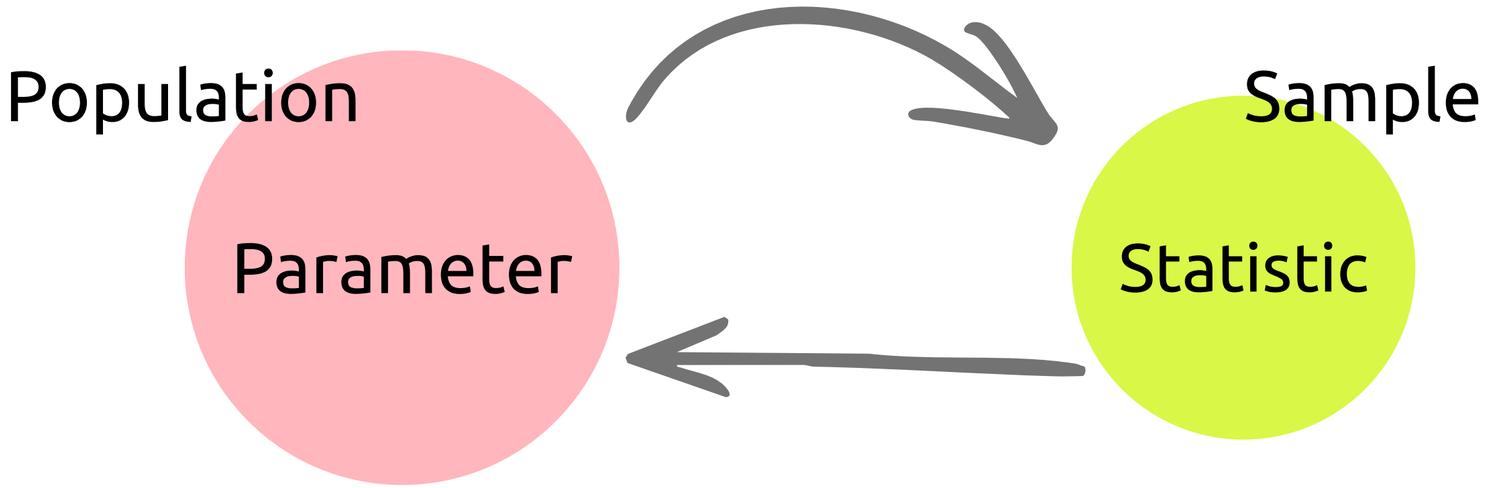
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WHAT IS STATISTICAL INFERENCE?

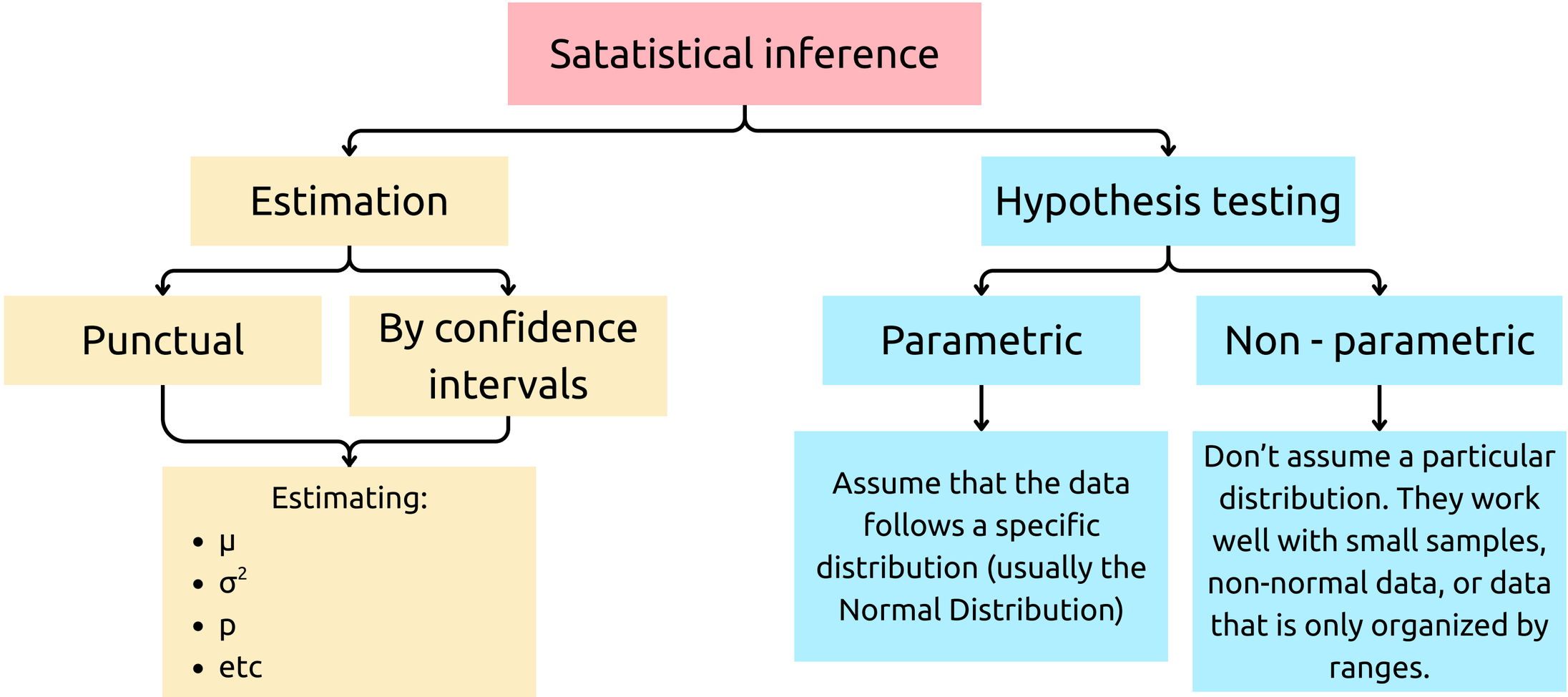
Inference

Branch of statistics that uses a set of methods or techniques to ***draw conclusions*** about a ***population's parameter***, from the information obtained in the ***sample***, called statistic.



Inference

The techniques of statistical inference are divided into two principal areas:



Johnson LL, Borkowf CB, Albert PS. An introduction to biostatistics: Randomization, hypothesis testing, and sample size estimation. En: Gallin JI, Ognibene FP, editores. Principles and practice of clinical research. 2.ª ed. Bethesda, MD: Academic Press; 2007. p. 181-198.

HYPOTHESIS TESTING

What is a hypothesis in statistics?

It is an ***assertion to be verified*** that is made about the ***numerical characteristics*** (parameters) or ***shape of the distribution*** (probability function) of one or more ***populations***.

How do we verify this?

Hypothesis testing: It is a procedure, based on sample evidence and probability theory, that is used to determine whether the hypothesis is a reasonable claim and should not be rejected or if it is not reasonable and should be rejected

Simple day to day examples:

- The time you spend in the Transmilenio follows a Normal Distribution



- The 20% of your friends have an iPhone



- The average wage of a worker in the construction sector is $785,500.



Null and alternative hypothesis

We will always need to avaluate these two possible scenarios:

H0

Null hypothesis

A statement about the value of the population. It implies the sign "="

H1

Alternative hypothesis

A statement that implies the opposite of the null hypothesis. It involves using the sign "≠"

Johnson LL, Borkowf CB, Albert PS. An introduction to biostatistics: Randomization, hypothesis testing, and sample size estimation. En: Gallin JI, Ognibene FP, editores. Principles and practice of clinical research. 2.ª ed. Bethesda, MD: Academic Press; 2007. p. 181-198.

VARIANCE ANALYSIS

ANOVA (ANalysis Of VAriance)

- Compares ***means*** across 3+ ***independent groups***
- Parametric
- Assumes: normality, equal variances (homoscedasticity)
- H_0: $\mu_1 = \mu_2 = ... = \mu_k$ (all group means are equal)
- H_1: At least one group mean differs
- Reject H_0 if F > F critical (or p-value < α)

We define the statistic as follows:

F = MSB / MSW

= (SSB/(k−1)) / (SSW/(N−k))

F is our statistic. If F is bigger then differences between the groups are much larger than internal "noise”

MSB: Variability between groups. How far the averages of each group are from the overall average.

MSW: It measures how much the individuals within a group vary from one another.

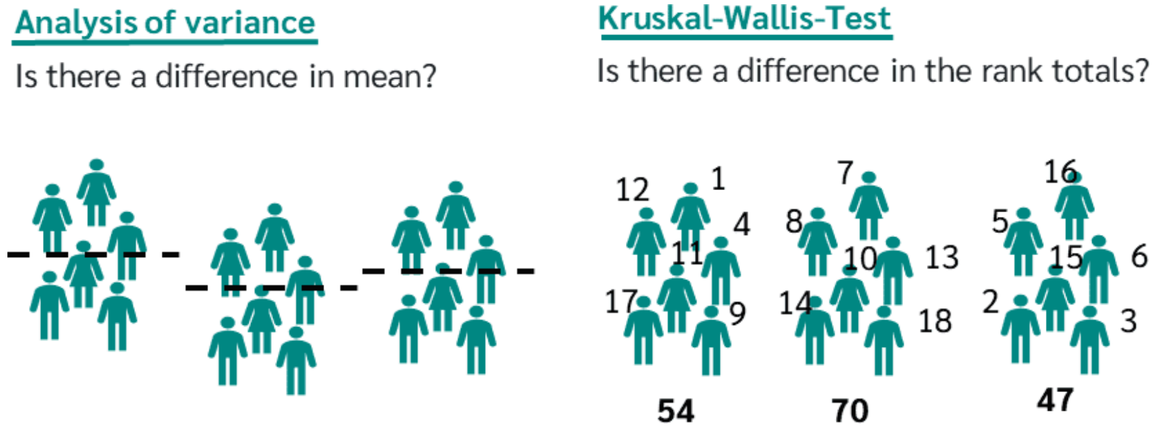
Kruskal - Wallis Test

- ***Non-parametric alternative*** to one-way ANOVA
- ***Compares medians / rank distributions*** of 3+ independent groups
- Does NOT assume normality, uses ranks of data
- H_0: All groups have the same distribution (equal medians)
- H_1: At least one group has a different distribution
- Reject H_0 if $H > x^2(k-1)$ critical or p-value < α

We define the statistic as follows:

$$H = [12 / N(N+1)] \cdot \Sigma(R_i^2 / n_i) - 3(N+1)$$

The H-statistic approximately follows a Chi-square distribution with (k–1) degrees of freedom for large samples.



Xia Y. Correlation and association analyses in microbiome study integrating multiomics in health and disease. Progress in Molecular Biology and Translational Science. 2020;171:309-491.

CORRELATION

Pearson Coefficient

- Measures linear relationship between two continuous variables
- Parametric
- Assumes: normality, linearity, homoscedasticity, no outliers
- H_0: $\rho = 0$ (no linear correlation in population)
- H_1: $\rho \neq 0$ (two-tailed) or $\rho > 0$ / $\rho < 0$ (one-tailed)
- Test statistic t follows a t-distribution

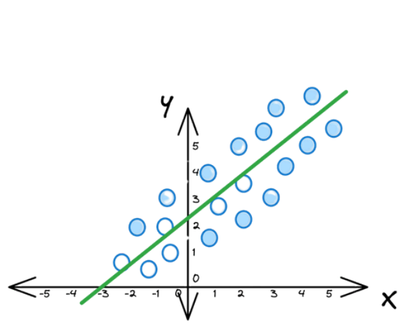
We define the statistic as follows

$$r = \Sigma(x_i - \bar{x})(y_i - \bar{y}) \,/\, \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot \Sigma(y_i - \bar{y})^2]}$$

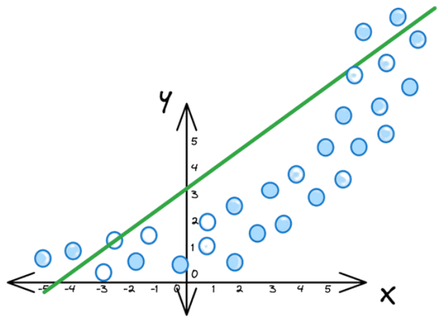
$$t = r\sqrt{(n-2)} \,/\, \sqrt{(1-r^2)}$$

Under H_0, the t-statistic follows a t-distribution with (n–2) degrees of freedom.

Pearson Correlation



Linear Data



Non-Linear Data

Kendall - Tau

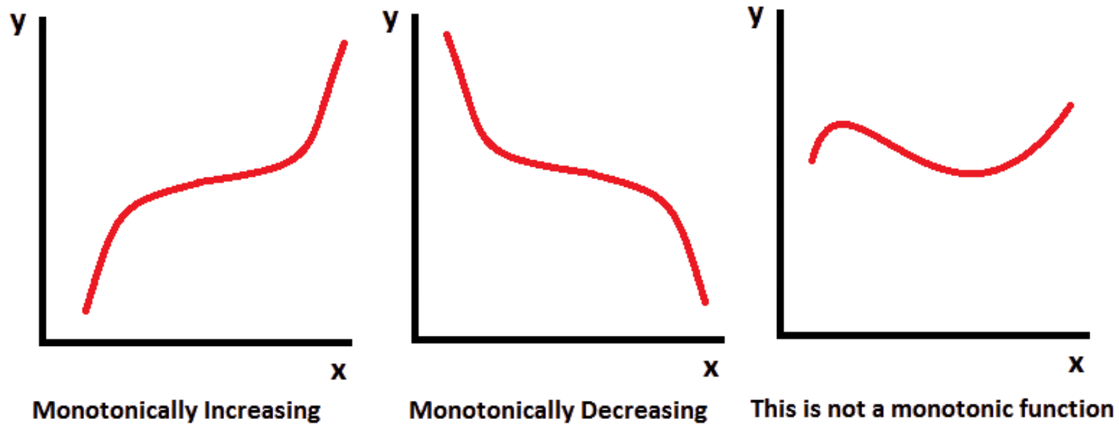
- Non-parametric measure of ordinal association between two variables
- Based on concordant (C) and discordant (D) pairs
- More robust than Pearson to outliers and non-normality
- H_0: $\tau = 0$ (no monotonic association)
- H_1: $\tau \neq 0$ (association exists)

We define the statistic as follows

$$\tau = (C - D) / [n(n-1)/2]$$

$$SE(\tau) = \sqrt{[2(2n+5) / (9n(n-1))]}$$

C= number of point pairs that follow the same trend.
D= number of pairs that contradict each other. While in one variable the value rises, in the other it falls.



University of Virginia Library. Correlation: Pearson, Spearman, and Kendall's tau [Internet]. Charlottesville (VA): University of Virginia; [citado el 12 de mayo de 2026]. Disponible en: https://library.virginia.edu/data/articles/correlation-pearson-spearman-and-kendalls-tau

FREQUENCY ANALYSIS

The Chi-Square Test (x^2)

- Tests association between two categorical variables (independence) or observed vs expected frequencies
- Requires: expected frequencies ≥ 5 in each cell (rule of thumb)
- H_0: Variables are independent (no association)
- H_1: Variables are NOT independent (association exists)
- Reject H_0 if $x^2 > x^2(df)$ critical or p-value < α

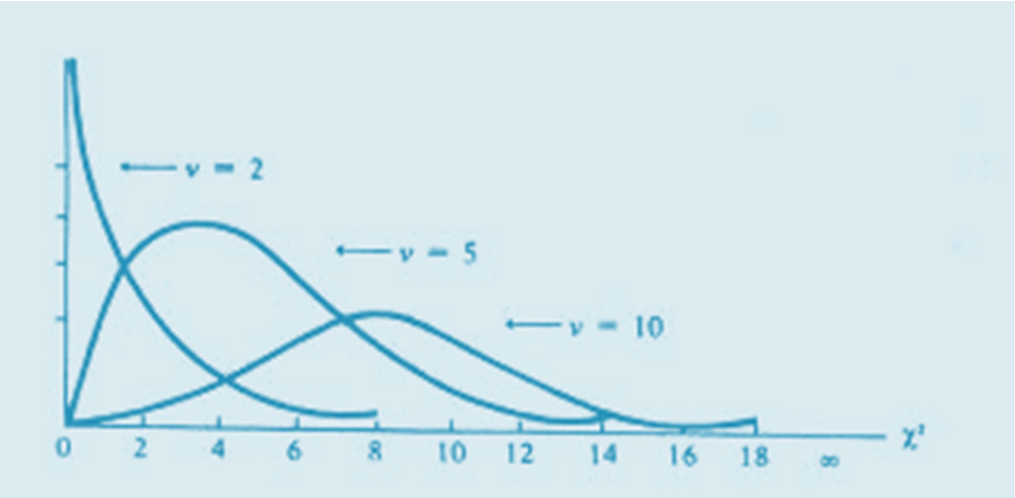
We define the statistic as follows

$$x^2 = \Sigma \left[(O_i - E_i)^2 / E_i \right]$$

$$E_i = (\text{Row Total} \times \text{Col Total}) / N$$

O_i = observed, E_i = expected

Depends on the degrees of freedom



McHugh ML. The chi-square test of independence. Biochem Med (Zagreb). 2013;23(2):143-9. doi: 10.11613/bm.2013.018. PMID: 23894860; PMCID: PMC3900058.

Thanks !!!!!